

SUPPLEMENTARY MATERIALS

Risk Assessment of Uterine Lesions (RAUL): a three-tier stratification model to guide fertility-preserving management.

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February 7, 2026

Abstract

This Supplementary Appendix provides a technical description of the analytical procedures used to develop, evaluate, and combine the three component classifiers that form the RAUL (Risk Assessment of Uterine Lesions) criterion. The document includes a detailed presentation of the dataset, the mathematical definition of the UMG index, and the construction of each classifier: a linear UMG-LDHtot boundary (Classifier 1), a geometric UMG-LDH5 partition (Classifier 2), and a simulation-based local probability estimator (Classifier 3). For each classifier, the rationale, computational implementation, graphical representation, and diagnostic performance metrics are reported. The final integrated RAUL score is described together with its confusion matrix, binary diagnostic properties, and likelihood ratios. All datasets, R scripts, and a standalone RAUL calculator are provided in a public GitHub repository to ensure full transparency and reproducibility.

1 Overview of the RAUL Supplementary Material

This Supplementary Appendix provides a fully reproducible description of the analytical workflow used to develop the RAUL (Risk Assessment of Uterine Lesions) criterion. As a companion to the main manuscript, which focuses on the clinical motivation, methodological summary, and diagnostic implications of the model, the present document details the full computational framework underlying the three component classifiers and their integration into the final three-tier RAUL index.

The Supplementary Material is structured to guide the reader through the entire process, beginning with a description of the dataset, followed by a step-by-step presentation of each classifier:

- **Classifier 1**, which combines the UMG index with total LDH separated by a linear boundary;
- **Classifier 2**, which partitions the UMG–LDH5 plane into geometric regions representing low, intermediate, and high risk patients;
- **Classifier 3**, which estimates a local probability of malignancy using a simulation-based probabilistic procedure.

For each classifier, we provide the underlying rationale, mathematical definition, graphical representation, and diagnostic performance metrics (sensitivity, specificity, predictive values, accuracy, and likelihood ratios when appropriate). The final section presents the construction and evaluation of the composite RAUL score (A/B/C), including its ability to identify a “no-risk” population with the highest sensitivity and negative predictive value.

All R code required to reproduce the analyses and figures is provided, together with links to the publicly available dataset and to a standalone RAUL calculator. These resources are hosted in an open GitHub repository, <https://github.com/umgstat/raul>, to ensure direct applicability to clinical or research settings.

2 Dataset Description

The analyses presented in this Supplementary Appendix are based on the full preoperative dataset used to construct and validate the RAUL score. The dataset comprises a total of 2453 women who underwent surgical treatment for a uterine mass between 2004 and 2016 at two tertiary gynecologic units

in Southern Italy. For each patient, the final histological diagnosis (benign or malignant) is available and serves as the reference standard for all diagnostic evaluations. Each row of the dataset corresponds to a single patient, and each column represents a clinical, biochemical, or algorithm-derived variable. The variables directly observed in the clinical setting are listed below:

- age: patient age at the time of surgery (years)
- outcome: final histopathological diagnosis, coded as benign or malignant
- LDHtot: total serum lactate dehydrogenase (U/L)
- LDH1, LDH2, LDH3, LDH4, LDH5: serum LDH isoenzymes (expressed in percentages)

These variables represent the original clinical and biochemical information used as the foundation for the RAUL algorithm. In addition to these observed variables, the dataset also contains several derived fields produced during the development of the three classifiers:

- UMG: original UMG risk index
- classifier1 and class1: continuous score and discrete class generated by Classifier 1
- class2: geometric risk class generated by Classifier 2
- p: local empirical probability of malignancy generated by Classifier 3
- class3: discrete class derived from the probability thresholds applied to Classifier 3
- raul: final three-level RAUL classification (A, B, C)

The dataset is fully anonymized and contains no patient identifiers. All variables are provided exactly as used in the analyses described in this document. The entire dataset is publicly available in the project GitHub repository, in `.csv` and in `.ods` format.

3 Mathematical Definition of the UMG Index

The UMG index is a preoperative biochemical score originally developed to characterize metabolic alterations associated with uterine masses. It combines information from LDH isoenzymes, with particular emphasis on LDH1 and LDH3, which reflect different components of aerobic and anaerobic cellular metabolism.

The UMG index is computed using the following formula:

$$UMG = LDH3 + \frac{24}{LDH1} \quad (1)$$

In this expression, LDH3 contributes directly to the index, while LDH1 enters through an inverse term, thereby giving greater weight to reduced LDH1 values. This formulation reflects the observation that malignant uterine lesions often present with elevated LDH3 activity together with a relative depletion of LDH1.

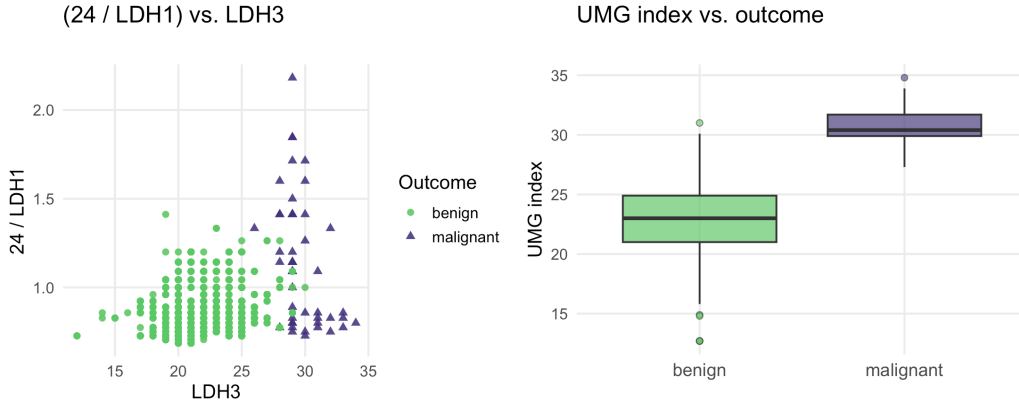


Figure 1: (*Left panel.*) Viridis palette scatter plot of LDH3 versus 24/LDH1 for all patients. Each point represents an individual observation, with benign lesions shown as green circles and malignant lesions as dark purple triangles. The variable 24/LDH1 corresponds to the inverse LDH1 term used in the definition of the UMG index. The plot illustrates the distinct distributions of benign and malignant cases in this biochemical space, with malignant lesions tending to show higher LDH3 values and increased 24/LDH1 compared with benign lesions. (*Right panel.*) Boxplot of the UMG index stratified by final histopathological outcome. Benign lesions are shown on the left and malignant lesions on the right. The plot displays the central tendency and variability of UMG values in each diagnostic category, and highlights the relative upward shift in UMG among malignant cases. This figure provides a compact visual summary of the discriminative behaviour of the UMG index between benign and malignant uterine lesions.

The UMG index is used as a continuous variable in the RAUL algorithm and serves as the core biochemical component of Classifier 1. All UMG values in the dataset were computed exactly using the formula above, with no transformations or rescaling applied, rounding to the first decimal.

Listing 1: import dataset in R and check UMG index

```
1 set.seed(123)
2 www = "https://raw.githubusercontent.com/umgstat/raul/
   refs/heads/main/dataset.csv"
3 rauldataset = read.csv(www, header = TRUE,
   stringsAsFactors = TRUE)
4 attach(rauldataset)
5 str(rauldataset)
6
7 all.equal(UMG, LDH3 + 24/LDH1, tolerance = 0.01)
```

4 Classifier 1: UMG and Total LDH Linear Boundary

4.1 Rationale

Classifier 1 integrates the UMG index with total serum LDH (LDHtot) to capture combined biochemical alterations associated with uterine sarcoma. The rationale behind this classifier is that UMG and LDHtot follow different statistical density distributions in benign versus malignant lesions. Benign masses tend to cluster along a lower and more compact range of LDHtot values for any given UMG value, whereas malignant lesions tend to shift upward in the UMG direction or outward toward higher LDHtot values.

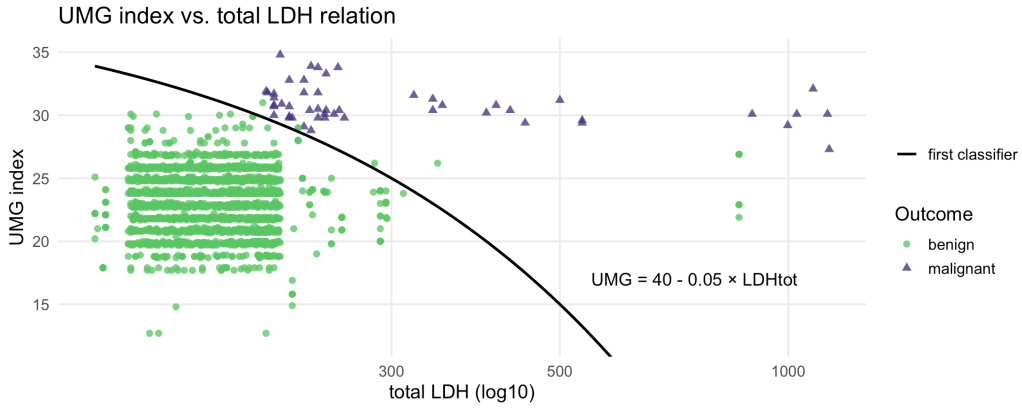


Figure 2: Scatter plot of the UMG index versus total LDH (LDHtot), shown on a logarithmic x-axis. Each point represents a single patient, with benign lesions displayed as green circles and malignant lesions as dark purple triangles. The solid black line corresponds to the decision boundary defined as $UMG = 40 - 0.05 \times LDH_{tot}$, which appears curved due to the logarithmic scaling of the LDHtot axis. Patients located above this boundary are assigned to high-risk Class C, those below to low-risk Class A, and - eventually - points lying exactly on the boundary correspond to Class B. The plot illustrates that nearly all malignant cases fall above the decision boundary, whereas benign cases predominantly lie below it, supporting the discriminative performance of Classifier 1.

This separation suggests that a simple linear boundary in the UMG–LDHtot plane can serve as an effective first discriminator. Classifier 1 assigns each patient to one of three classes (A, B, or C) depending on the position relative to this line.

4.2 Formula

The decision boundary is defined by the equation:

$$UMG = 40 - 0.05 \times LDH_{tot} \quad (2)$$

Patients are classified according to the following rules:

- Class A (low risk): $UMG < 40 - 0.05 \times LDH_{tot}$
- Class B (intermediate): $UMG = 40 - 0.05 \times LDH_{tot}$
- Class C (high risk): $UMG > 40 - 0.05 \times LDH_{tot}$

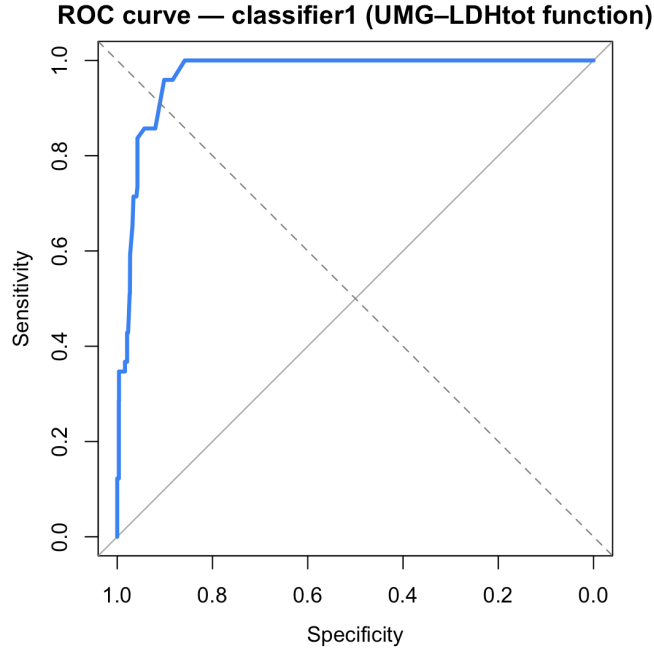
4.3 Diagnostic Performance

When Class C is considered a positive test (high risk) and Class A a negative test (low risk), Classifier 1 achieves the following diagnostic metrics (the AUC value derives from the continuous `classifier1` score. CI values represent exact or DeLong 95 percent confidence intervals):

Outcome	Class A	Class C
Benign	2391	13
Malignant	0	49

Measure	Estimate	95% CI
Sensitivity	1.000	[0.927–1.000]
Specificity	0.995	[0.991–0.997]
PPV	0.790	[0.668–0.883]
NPV	1.000	[0.998–1.000]
Accuracy	0.995	[0.991–0.997]
AUC (continuous <code>classifier1</code>)	0.969	[0.959–0.979]

The first classifier therefore reveals to provide the highest sensitivity and negative predictive value, meaning that no malignant lesion was ever assigned to Class A. The specificity appears to be also very high, with only a small fraction of benign lesions lying above the decision boundary.



5 Classifier 2: UMG–LDH5 Triangular Rule

5.1 Rationale

Classifier 2 is based on the joint behaviour of the UMG index and the LDH5 isoenzyme, which is known to reflect enhanced glycolytic activity observed in malignant cells as part of the Warburg effect. Malignant uterine lesions tend to exhibit elevated LDH5 values for a given UMG level, whereas benign lesions remain confined within a more restricted metabolic region. This motivates a geometric approach to risk stratification in the UMG–LDH5 plane.

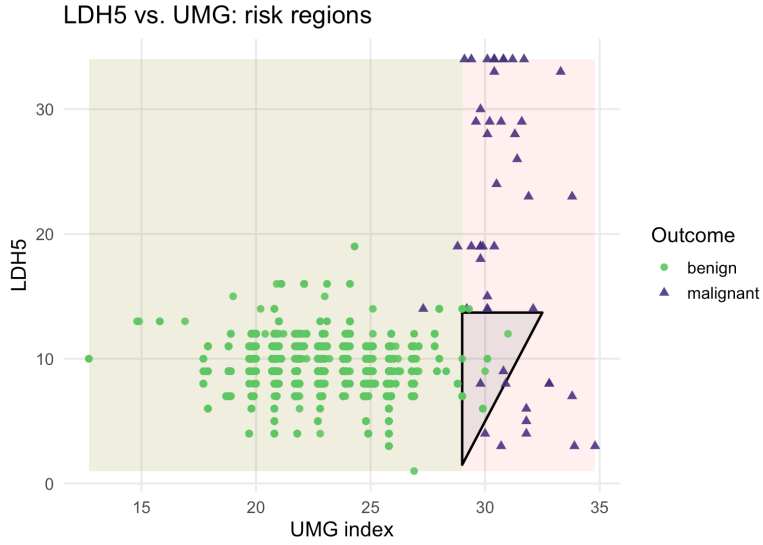
Classifier 2 partitions the space into three risk classes (A, B, and C) using a combination of UMG thresholds, LDH5 thresholds, and a well-defined triangular intermediate-risk region. This construction allows separation of clearly benign patterns, clearly malignant patterns, and a transitional region of partial overlap.

5.2 Geometric Construction

The following rules define the three classes:

- Class A (low risk): $UMG < 29$. This region appears on the left side of the plot and is coloured green. In the observed data, no malignant cases fall below this threshold.

- Class C (high risk): $UMG \geq 29$ and $LDH5 > 13.7$, or $UMG \geq 29$ and $LDH5 < 3.5 * UMG - 100$. These two logical conditions identify high-risk regions above and below the intermediate area. This region is coloured red.
- Class B (intermediate risk): the triangular region with vertices (29, 1.5), (29, 13.7), and (32.5, 13.7). This region is coloured grey.



5.3 Diagnostic Performance

Outcome	Class A	Class B	Class C
Benign	2383	18	3
Malignant	2	13	34

Here above the contingency table for Classifier 2 is reported. The three classes represent low-risk (Class A), intermediate-risk (Class B), and high-risk (Class C) regions defined in the UMG–LDH5 plane. Values indicate the absolute frequencies of benign and malignant lesions assigned to each class. When Class A is considered a negative test (low risk) and Classes B and C are grouped as positive (non-low-risk), the classifier achieves good sensitivity and higher negative predictive value. Nevertheless, two malignant cases were placed outside the risky region, and benign misclassifications appears to be placed near the triangular region.

Measure	Estimate	95% CI (low)	95% CI (high)
Sensitivity	0.959	0.860	0.995
Specificity	0.991	0.987	0.995
PPV	0.691	0.567	0.798
NPV	0.999	0.997	1.000
Accuracy	0.991	0.986	0.994

6 Classifier 3: Simulation-Based Probability Estimator

6.1 Rationale

Classifier 3 adopts a fundamentally different approach compared with the first two classifiers. Instead of relying on fixed geometric boundaries, it estimates the local probability that a given point belongs to a benign or a malignant region in the UMG–LDH5 plane. This is achieved through a simulation-based frequentist procedure, which generates two synthetic point clouds representing the benign and malignant metabolic distributions.

The key idea is that the density of simulated benign points and simulated malignant points in the local neighborhood of a patient-specific coordinate (UMG, LDH5) can be used to approximate the probability that the observed point belongs to either distribution. This approach allows flexible modeling of regions where the benign and malignant populations overlap, without imposing explicit linear or geometric constraints.

6.2 Simulation Procedure

Two point clouds of size $n = 1000$ each are generated:

- **Benign cloud:** The benign distribution is approximated using a bivariate normal model. The mean values, standard deviations, and correlation between UMG and LDH5 are computed from the benign subset of the dataset. Using these empirical parameters, a normal bivariate variable is generated in order to preserve and to reproduce the observed correlation structure. The resulting coordinates (xB, yB) form the simulated benign ('green') cloud.
- **Malignant cloud:** The malignant distribution is approximated using a wide bivariate uniform distribution. This choice reflects the broader and more irregular dispersion observed among malignant cases. UMG values are drawn uniformly from the interval 29 to 40, while LDH5

values are drawn uniformly from the interval 1 to 40. The resulting coordinates (xM, yM) form the simulated malignant ('purple') cloud.

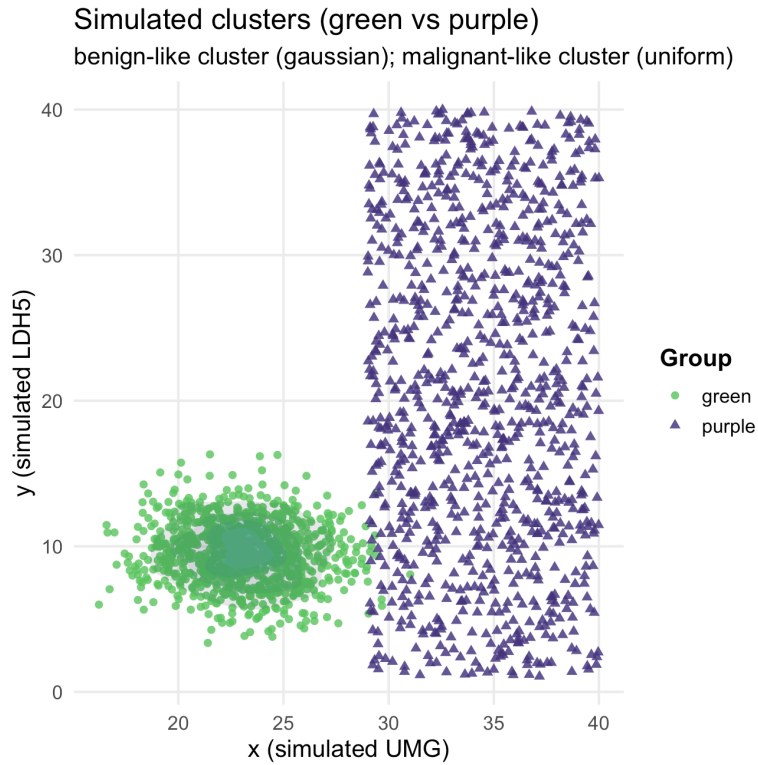


Figure 3: Simulated benign-like and malignant-like point clouds in the UMG–LDH5 plane. The green cluster is generated from a bivariate normal distribution whose empirical mean, variance, and correlation match those observed in the benign subset of the real dataset. The purple cluster is generated from a wide bivariate uniform distribution reflecting the broader and more irregular dispersion exhibited by malignant lesions. Each point corresponds to one simulated sample.

The two sets of points are combined into a single synthetic reference population containing both benign and malignant labels.

Listing 2: The two sets of synthetic points

```

1 n=1000
2
3 xbenign = UMG[outcome == "benign"]
4 ybenign = LDH5[outcome == "benign"]
5 mx = mean(xbenign); sx = sd(xbenign);
6 my = mean(ybenign); sy = sd(ybenign);
7 r = cor(xbenign, ybenign)

```

```

8
9 n1=rnorm(n); n2=rnorm(n)
10 n3=r*n1+sqrt(1-r^2)*n2
11 xB = mx+sx*n1
12 yB = my+sy*n3
13
14 xM = runif(n, 29, 40)
15 yM = runif(n, 1, 40)
16
17 x = c(xB, xM)
18 y = c(yB, yM)
19 colour = factor(c(rep("green", n), rep("purple", n)))

```

6.3 Probability Calculation

For any patient with coordinates (x_{test}, y_{test}) , the Euclidean distance to each simulated point is computed:

$$d_i = \sqrt{(x_i - x_{test})^2 + (y_i - y_{test})^2}.$$

A local neighborhood of (x_{test}, y_{test}) is defined by selecting the radius corresponding to the 10th percentile of all distances:

$$r = \text{quantile}(d, 0.10).$$

All simulated points falling within this radius are considered local neighbors. Let:

- n_B = number of benign simulated points inside the circle
- n_M = number of malignant simulated points inside the circle

The empirical probability of malignancy is then defined as:

$$p(x_{test}, y_{test}) = \frac{n_M}{n_B + n_M}.$$

Listing 3: example of probability estimate

```

1 xtest = 28.5
2 ytest = 13.5
3 distance = sqrt( (x-xtest)^2 + (y-ytest)^2 )
4 radius = quantile(distance, 0.1)
5 nearest = which(distance < radius)

```

```

6 table(colour[nearest])
7 (p = table(colour[nearest]][[2]] / sum(table(colour[
  nearest]))))

```

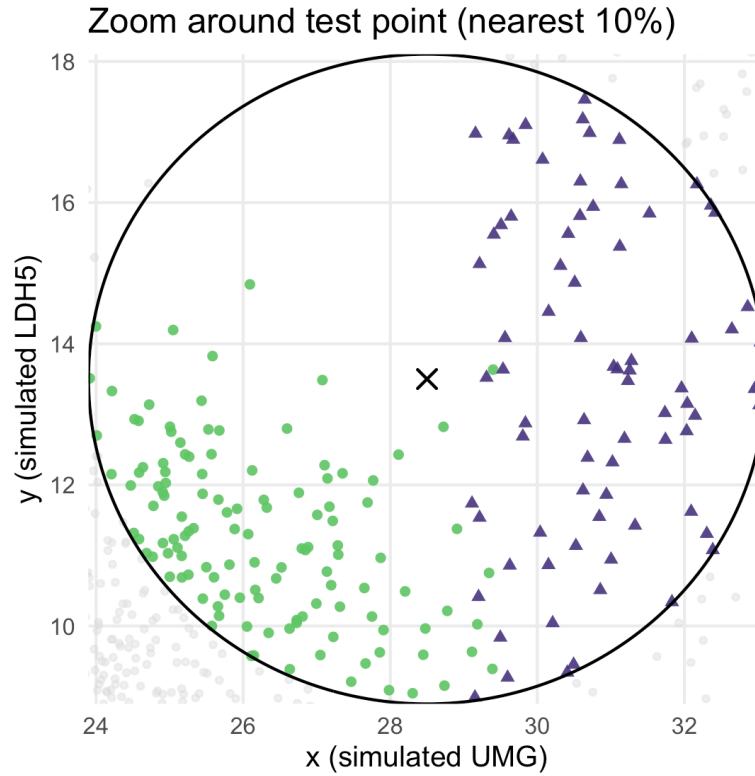


Figure 4: Local neighbourhood used for the probability computation in Classifier 3. The black cross marks the test point $(x_{\text{test}}, y_{\text{test}})$ for which the local malignant probability $p(x, y)$ is estimated. Distances from this point to all simulated benign-like and malignant-like samples are computed, and the radius of the circle corresponds to the 10th percentile of all distances. Points inside the circle (highlighted with their original green or purple colours) represent the nearest neighbours used to estimate the empirical probability $p(x, y) = n_M / (n_B + n_M)$. The circle therefore defines the adaptive local region in which benign and malignant densities are compared. This plot illustrates the core mechanism of the simulation-based probability estimator: as an example, a patient with $\text{UMG} = 28.5$ and $\text{LDH5} = 13.5$ might yield $n_B = 124, n_M = 76, p = 0.38$.

6.4 Class Assignment

The continuous probability $p(x, y)$ is converted into a three-class categorical variable using the following thresholds:

- Class A (low risk): $p < 0.01$
- Class B (intermediate risk): $0.01 \leq p < 0.15$
- Class C (high risk): $p \geq 0.15$

The lower threshold for Class A ($p < 0.01$) was intentionally chosen to be extremely conservative. Uterine sarcoma is a rare disease, representing only a small fraction of uterine masses encountered in routine gynecologic practice: the main priority of Classifier 3 is to avoid any false negative assignments. For this reason, only points with an exceptionally low local malignant probability are allowed to enter Class A. This guarantees strong rule-out performance and reflects the clinical requirement that patients classified as low risk should have virtually no chance of underlying sarcoma.

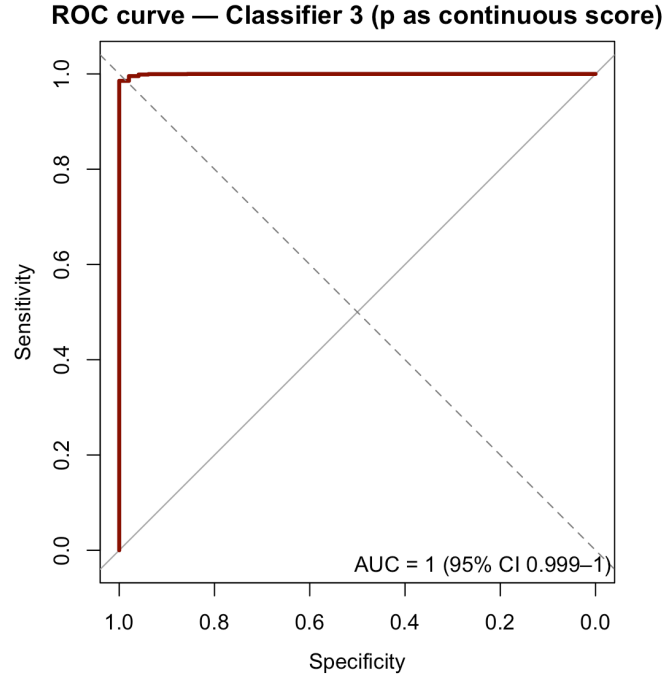
6.5 Diagnostic Performance

Using Class A as a negative test (low risk) and Classes B and C as a positive test (non-low-risk), the resulting diagnostic metrics are:

Measure	Estimate	95% CI
Sensitivity	1.000	[0.927–1.000]
Specificity	0.924	[0.913–0.935]
PPV	0.212	[0.161–0.271]
NPV	1.000	[0.998–1.000]
Accuracy	0.926	[0.915–0.936]
AUC (continuous p)	1.000	[0.999–1.000]

Table 1: Diagnostic performance of Classifier 3. Binary metrics refer to the A vs (B+C) dichotomisation. The AUC is computed using the continuous probability p with DeLong 95% confidence intervals.

Classifier 3 therefore achieves perfect sensitivity and perfect negative predictive value. All malignant lesions were assigned to high-risk Class C. Benign misclassifications reflect the natural overlap of metabolic profiles in regions where simulated malignant density exceeds simulated benign density. Despite modest precision, the classifier provides an excellent rule-out instrument and complements the geometric classifiers by capturing nonlinear patterns of local density.



7 Final RAUL Score

7.1 Construction

The final RAUL score integrates the outputs of the three individual classifiers described in the previous sections. Each classifier assigns patients to one of three categories (A, B, C), corresponding to low, intermediate, and high risk. The final RAUL class is obtained by combining the three intermediate outputs and assigning each patient to the final risk class, according to the rule exposed in our research paper:

- all three classifiers A $\Rightarrow$ RAUL Class A
- two or three classifiers C $\Rightarrow$ RAUL Class C
- all remaining combinations $\Rightarrow$ RAUL Class B

7.2 Final Contingency Table

The following table reports the distribution of benign and malignant lesions across the three RAUL classes:

Outcome	Class A	Class B	Class C
Benign	2216	184	4
Malignant	0	0	49

All malignant lesions (49 out of 49) were assigned to RAUL class C, and none were allocated to class A or class B. Among benign cases, 2216 were declared as class A, null risk, and only 4 were placed in class C.

7.3 Binary Evaluation (A vs B+C)

For clinical interpretation, the RAUL score can be dichotomised by considering class A as a negative test (no risk) and classes B and C as a positive test (non-low-risk). The corresponding confusion matrix is:

Outcome	Negative	Positive
Benign	2216	188
Malignant	0	49

From this table, the diagnostic metrics are:

- Sensitivity: 1.000
- Specificity: 0.922
- Positive Predictive Value: 0.207
- Negative Predictive Value: 1.000
- Accuracy: 0.923

The RAUL score achieves perfect sensitivity and perfect negative predictive value: no malignant lesion was ever classified as class A. This property confirms the ability of RAUL to define a reliable no-risk population. The specificity remains high, with only 7.8 percent of false positive, i.e. benign lesions assigned to a non-low-risk category.

7.4 Likelihood Ratios

Likelihood ratios were computed for the binary RAUL test (A vs B+C):

- Positive likelihood ratio (LR+): 12.79
- Negative likelihood ratio (LR-): 0.00

An LR- of zero enhances the fact that the RAUL score is able to implement a perfect rule-out screening capability, as no malignant cases were assigned to the negative (no-risk) category. An LR+ of 12.79 indicates that a positive RAUL score is strongly associated with malignancy and provides useful rule-in information.

8 Reproducibility Resources

To ensuring transparency and reproducibility, all data, R scripts, and supporting tools used in this study are publicly available online. This section provides a detailed overview of these resources.

8.1 R Script

A complete R script is provided in the GitHub repository associated with this study. The script reproduces all analyses reported in the Supplementary Appendix, including:

- computation of the UMG index
- construction of Classifier 1, Classifier 2, and Classifier 3
- generation of all figures (UMG–LDHtot plot, UMG–LDH5 geometric plot, simulation-based probability plots)
- calculation of diagnostic performance metrics
- production of the final RAUL classification

All numerical results and graphical outputs appearing in this document can be regenerated exactly by running the script without modification.

8.2 Data Availability

The full anonymized dataset of 2,453 patients is available as a comma separated values (CSV) and open data (ODS) file in the public GitHub repository. The dataset contains all clinical, biochemical, and algorithm-derived variables used in the construction and evaluation of the RAUL score. No personal identifiers are included, and the dataset complies with applicable privacy standards.

8.3 RAUL Calculator

To facilitate the practical use of the RAUL score in clinical and research settings, a standalone RAUL calculator is provided in the form of an .ODS spreadsheet. This tool allows users to input LDHtot, LDH1, LDH2, LDH3, LDA4 and LDH5 values, and automatically computes:

- UMG

- Classifier 1 output
- Classifier 2 output
- Classifier 3 (approximate) probability and class
- final RAUL class (A, B, or C)

The calculator may be used for replication of the study results, or to explore the behaviour of the RAUL algorithm in hypothetical clinical scenarios. It is fully consistent with the R code and with the algorithms described in this Supplementary Appendix.



Figure 5: the RAUL calculator as a Desktop .ODS spreadsheet

8.4 Repository Access

All reproducibility resources, including the dataset, scripts, and calculator, are permanently hosted at:

<https://github.com/umgstat/raul>